

CUSTOMER BEHAVIOR ANALYSIS REPORT

1. INTRODUCTION

Customer behavior analysis is an important part of business analytics that helps organizations understand how customers interact with products and services.

In this project, we analyze a real-world customer transaction dataset to study purchasing patterns, product preferences, payment behavior, and customer churn.

The main goal is to extract meaningful insights that can help businesses improve customer retention, increase revenue, and optimize marketing strategies.

2. OBJECTIVE OF THE PROJECT

The main objectives of this analysis are:

- To clean and preprocess raw customer transaction data**
- To perform exploratory data analysis (EDA)**
- To understand customer purchasing behavior**
- To identify top-performing product categories**
- To analyze customer churn patterns**
- To study payment method preferences**
- To identify high-value customers**
- To generate actionable business insights**

3. DATASET DESCRIPTION

The dataset contains detailed customer transaction records with the following features:

- Customer ID: Unique identifier for each customer**
- Customer Name: Name of the customer**
- Age / Customer Age: Age of customer**
- Gender: Gender of customer**
- Purchase Date: Date and time of transaction**
- Product Category: Type of product purchased**
- Product Price: Price per product**

- **Quantity:** Number of items purchased
- **Total Purchase Amount:** Total transaction value
- **Payment Method:** Mode of payment used
- **Returns:** Whether product was returned
- **Churn:** Customer churn status (0 = No, 1 = Yes)

4. DATA CLEANING AND PREPROCESSING

Before analysis, the dataset was cleaned and prepared:

- **Checked for missing values** → No missing data found
- **Removed duplicate records** if any
- **Verified data types** for each column
- **Handled redundant columns** (Age vs Customer Age)
- **Ensured consistency** in categorical variables

This ensured that the dataset was ready for accurate analysis.

5. EXPLORATORY DATA ANALYSIS (EDA)

EDA was performed to understand customer patterns and trends.

5.1 Customer Demographics

- **Age distribution** was analyzed to understand customer segments
- **Gender distribution** showed the proportion of male and female customers

5.2 Product Category Analysis

- **Product categories** were analyzed to identify most purchased items
- **Certain categories** like Electronics and Home products showed higher demand

5.3 Payment Method Analysis

- **Payment methods** such as Credit Card and PayPal were most commonly used
- This shows preference for digital payment systems

5.4 Purchase Behavior

- **Total purchase amounts** were analyzed to identify spending patterns
- **Some customers** contributed significantly higher revenue

6. CUSTOMER CHURN ANALYSIS

Customer churn is a critical business metric.

Observations:

- **Majority of customers are retained (Churn = 0)**
- **A significant portion of customers (~20%) have churned**
- **Churned customers represent potential revenue loss Business Impact:**
- **High churn indicates need for better retention strategies**
- **Identifying churn patterns helps improve customer loyalty programs**

7. HIGH VALUE CUSTOMER ANALYSIS

A small group of customers contributes significantly to total revenue.

Findings:

- **Top customers generate a large portion of sales**
- **These customers should be prioritized for retention**
- **Loyalty programs can help retain these valuable customers**

8. KEY INSIGHTS

From the analysis, the following insights were derived:

- **Electronics and Home categories are the most purchased product types**
- **Credit Card and PayPal dominate payment methods**
- **Customer spending varies across different age groups**
- **A small number of customers contribute high revenue**
- **Customer churn is present and needs attention**
- **Purchase behavior is influenced by product category and age**

9. BUSINESS RECOMMENDATIONS

Based on the analysis, the following recommendations are suggested:

1. Improve Customer Retention

Focus on reducing churn by offering discounts, loyalty points, and personalized offers.

2. Target High-Value Customers

Identify and reward top customers with special benefits to increase loyalty.

3. Optimize Product Strategy

Increase focus on high-performing categories like Electronics and Home products.

4. Improve Marketing Strategy

Use customer age and behavior data to target specific customer groups.

5. Enhance Payment Experience

Promote preferred payment methods and ensure smooth transaction experience.

10.CUSTOMER SEGMENTATION (RFM ANALYSIS)

Customer segmentation was performed using RFM model:

- **Recency (how recent purchase was)**
- **Frequency (how often purchases occur)**
- **Monetary (total spending)**

Based on RFM scoring, customers were grouped into:

- **Champions**
- **Loyal Customers**
- **At Risk Customers**
- **Lost Customers**
- **Regular Customers**

This segmentation helps in identifying high-value and churn-risk customers.

11.CONCLUSION

This project successfully analyzed customer behavior using real-world data. The insights generated help understand customer preferences, spending behavior, and churn patterns.

These findings can help businesses make data-driven decisions to improve customer satisfaction, increase revenue, and reduce churn.